

DAMATH DE SILVA



BSC ENGINEERING HONOURS
COMPUTER ENGINEERING UNDERGRADUATE,
UNIVERSITY OF JAFFNA.

[GitHub](#) [LinkedIn](#) [Portfolio](#) damathdesilva@gmail.com Negombo, Sri Lanka +94773419799

PROFESSIONAL SUMMARY

I am a final-year Computer Engineering undergraduate at the University of Jaffna, with hands-on internship experience in AWS serverless backend development at WealthOS (acquired by JPMorgan Chase). Skilled in Node.js/TypeScript, cloud infrastructure (AWS), Infrastructure as Code (Terraform), and RESTful API design. Passionate about backend engineering, cloud-native architectures, and DevOps practices. Seeking a Software Engineering or DevOps role to contribute to scalable, production-grade systems.

PROFESSIONAL EXPERIENCE

Backend Developer Intern | [WealthOS \(acquired by JPMorgan Chase\)](#) Nov 2025 – May 2026
Employee Wellness Allowance System — AWS Serverless Platform

- Architected and delivered a full Employee Wellness Allowance System on AWS serverless infrastructure, enabling employees to submit, track, and manage reimbursement claims end-to-end.
- Built scalable backend services in Node.js (TypeScript) using Lambda, API Gateway, DynamoDB, S3, SES, EventBridge, Cognito, VPC, and Route 53; designed for high availability and low operational overhead.
- Implemented secure authentication and authorization with AWS Cognito and Google OAuth 2.0 integration, enforcing role-based access control across Super Admin, HR, and Finance roles.
- Designed a claim management workflow supporting prescription/invoice uploads (S3), claim lifecycle management (create/view/delete), and multi-tier approval routing with validations and audit controls.
- Provisioned and managed the entire AWS infrastructure using Terraform (IaC), ensuring reproducible, version-controlled deployments across environments.
- Wrote unit and integration tests with Mocha and Chai; used LocalStack for local AWS service simulation, reducing development feedback cycles significantly.

PROJECTS

1. **SHOPLANE (Full-Stack E-Commerce Platform)** Jan 2026 – Present

Tech: MongoDB, Express.js, React.js, Node.js, JWT, Tailwind CSS, React Icons, Postman

- Engineered a scalable e-commerce platform using the MERN stack with secure JWT authentication, role-based access control, and robust RESTful APIs for product, cart, and order management.
- Designed a fully responsive React.js storefront with dynamic interactions; validated all API endpoints through comprehensive Postman test suites.

2. **BOOKIFY – Booking and Management Platform (Team Project)** Jan 2025 - May 2025

Tech: Spring Boot, React.js, MySQL, JWT, Git & GitHub, Jira, Figma

- Led backend development and project management for a booking platform supporting one-time and recurring reservations, automated reminders, and analytics dashboards.
- Managed sprint planning and task tracking in Jira; designed system architecture and UI wireframes in Figma, delivering the project on schedule with a team of 5.

3. SMART WATER – IoT-Based Household Water Management System (Team Project)

Tech: ESP32, C, React.js, Firebase

Jan 2025 - May 2025

- Developed firmware in C for ESP32 to monitor water inflow, outflow, level, and TDS quality; integrated sensor data pipeline to Firebase with secure real-time transmission.

4. Optimizing AI Algorithms for Low-Power and Edge Devices in Resource-Constrained Environments (Final Year Research Project)

Jan 2025 - Present

Tech: Python, TensorFlow, Edge Inference, Embedded Systems

- Developed an agent-based Edge AI system for dynamic selection of optimized pretrained models on resource-constrained microcontrollers, supporting multimodal inference (image, audio, text, sensor).
- Implemented lightweight deployment strategies, achieving real-time inference on low-power edge devices.

TECHNICAL SKILLS

- **Cloud & DevOps:** AWS (Lambda, API Gateway, DynamoDB, S3, Cognito, SES, EventBridge, VPC, Route 53), Terraform (IaC), LocalStack, CI/CD
- **Languages:** TypeScript, JavaScript (Node.js), Python, Java, C++, PHP, SQL
- **Backend & APIs:** Express.js, RESTful APIs, JWT, OAuth 2.0, Serverless Architecture
- **Frontend:** React.js, HTML, CSS, Tailwind CSS
- **Databases:** DynamoDB, MySQL, MongoDB
- **Testing:** Mocha, Chai, Postman, Unit & Integration Testing
- **Tools & Platforms:** Git/GitHub, Jira, Figma, Linux
- **ML/AI (Familiar):** TensorFlow, Scikit-learn, OpenCV

EDUCATION

BSc Engineering (Hons) in Computer Engineering

Apr 2022 - Present

AWARDS & ACHIEVEMENTS

- 1st Place, YARL XTREME Hackathon (Oct 2024) 
- IEEE Xtreme 18.0 — Sri Lanka Rank: 116 (Oct 2024) 
- MoraXtreme 9.0 — 61st out of 300+ competing teams (Oct 2024) 
- CodeRally 5.0 — 12th out of 40 teams (Sep 2024) 

LEADERSHIP & VOLUNTEERING

Secretary - IEEE Robotics and Automation Society (RAS), University of Jaffna

Feb 2024 - Apr 2026

REFEREES

DR. (Mrs.) J. Jananie

Senior Lecturer Grade II

Department of Computer Engineering,

Faculty of Engineering,

University of Jaffna

jananie@eng.ifn.ac.lk

+94 70 490 4957

Mr. Y. Pirunthapan

Lecturer

Department of Computer Engineering,

Faculty of Engineering,

University of Jaffna

pirunthapany@eng.jfn.ac.lk

+94 21 206 0160